CoRoT-2b

R-1300

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_200920` · created 2026-07-31T17:12:06+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459112.7037 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1781 +/- 0.0093
Transit depth (Rp/Rs)^2	3.17 +/- 0.33 [%]
Orbital Inclination (inc)	90.0 +/- 1.07
Airmass coefficient 1 (a1)	1.041 +/- 0.017
Airmass coefficient 2 (a2)	-0.0262 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	1.64 %
Best Comparison Star	#3 - [529, 457]
Optimal Aperture	3.7
Optimal Annulus	9.95
Transit Duration (day)	0.0975 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1781 ± 0.0093 vs 0.1667 (deviation 1.17σ)
Duration fit vs published	0.0975 d vs 0.095 d (deviation 2.17σ)
Mid-time O–C	6.7 min
Depth SNR	18.3
Residual scatter	1.64 %

Light curve

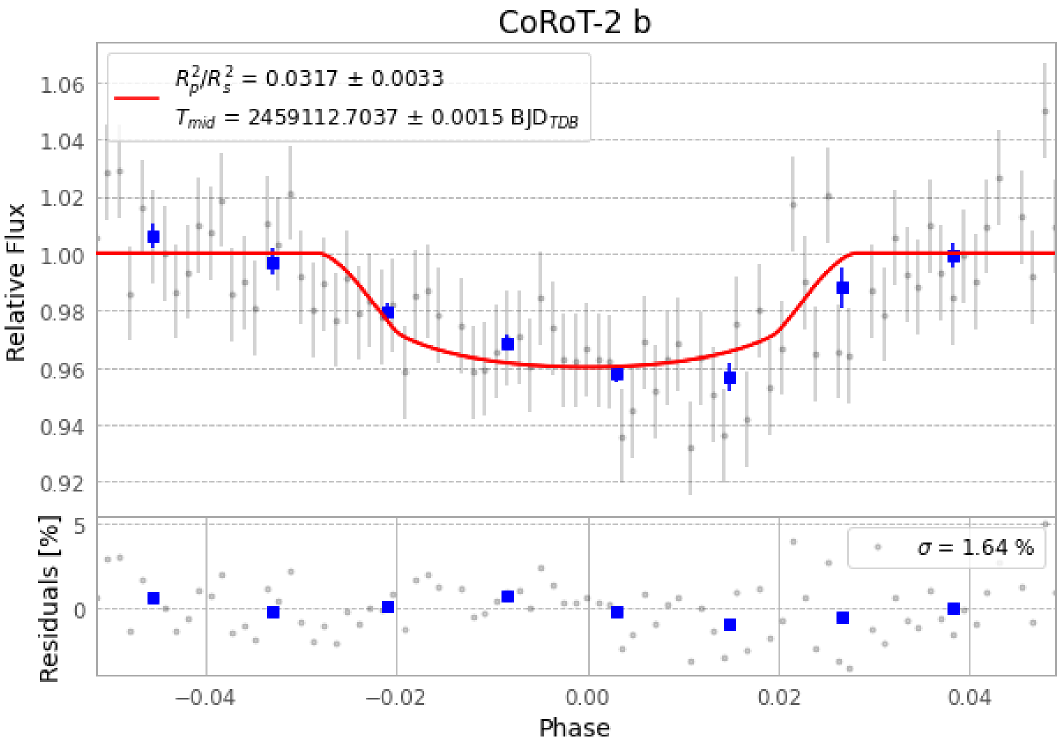

Figure 1 — FinalLightCurve_CoRoT-2 b_2020-09-19.png (SHA-256 038a67e285699075…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [140, 236], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f51a5197c912612f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

85 FITS frames (56 MB), `CoRoT-2200920023914.FITS` → `CoRoT-2200920065113.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-200920.log (3d3b6b45069e…), FinalLightCurve_CoRoT-2 b_2020-09-19.png (038a67e28569…), FinalLightCurve_CoRoT-2 b_2020-09-19.csv (225b837c600e…)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-31 17:12Z.